

No announcements

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5: Step 1: First, setup equation  $\nabla f = \lambda \nabla g$  and simplify.

$$f = gxe^y, \quad g = x^2 + y^2, \quad \text{given } g = 2.$$

$$g \langle e^y, xe^y \rangle = \nabla f = \lambda \nabla g = 2\lambda \langle x, y \rangle$$

$$\Rightarrow \begin{cases} ge^y = 2\lambda x \\ gxe^y = 2\lambda y \end{cases}$$

Step 2: Solve system of equations

$$\text{Notice } 2\lambda x^2 = gxe^y = 2\lambda y \Rightarrow$$

$$\lambda x^2 = \lambda y \Rightarrow x^2 = y \quad \text{or} \quad \lambda = 0.$$

If  $\lambda = 0$ , then plug in  $\Rightarrow ge^y = 0$ ,

so  $e^y = 0$ , no solutions.

$$\text{So } x^2 = y \Rightarrow 2 = x^2 + x^4 \Rightarrow$$

$$(x^2 - 1)(x^2 + 2) = 0, \text{ so}$$

$x^2 = 1, -2 \Rightarrow$  only  $x = \pm 1$ ,  
in either case  $y = 1$ . So  
 $(1, 1), (-1, 1)$  are candidates.

Step 3: Plug solutions back in  
to see which is min/max.

$$f(1, 1) = 9e, \quad f(-1, 1) = -9e,$$

So  $\min = -9e, \max = 9e$

(b)  $f = 81x^2 + y^2, \quad g = 4x^2 + y^2 = 9$

Step 1: Setup  $\nabla f = \lambda \nabla g$ .

$$2 \langle 81x, y \rangle = \nabla f = \lambda \nabla g = 2\lambda \langle 4x, y \rangle$$

$$\Rightarrow \begin{cases} 81x = 4\lambda x \\ y = \lambda y \end{cases}$$

Step 2: solve system.

Begin with  $y = \lambda y$ .  $\lambda = 1$  or  $y = 0$ .

If  $y=0$ : Then  $4x^2=9 \Rightarrow x^2=\frac{9}{4}$   
 $\Rightarrow f=81x^2+y^2=81x^2=\frac{729}{4}$

If  $\lambda=1$ : Plug into other equation

$$\Rightarrow 81x=4x \Rightarrow x=0.$$

Then  $9=9=y^2 \Rightarrow f=y^2=9.$

Step 3: Compare values found

We only found  $9$  &  $\frac{729}{4}$ .

So  $\min=9, \max=\frac{729}{4}$

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3: Find abs max & min

(2)  $f=x^2+y^2+x^2y+9, D=\{x, y \mid |x|, |y| \leq 1\}$

Step 1: Solve  $\nabla f=0$  since any extrema in interior must be a critical point, and any critical point has  $\nabla=0$  when  $\nabla$  defined.

$$\vec{0} = \nabla f = \langle 2x + 2xy, 2y + x^2 \rangle$$

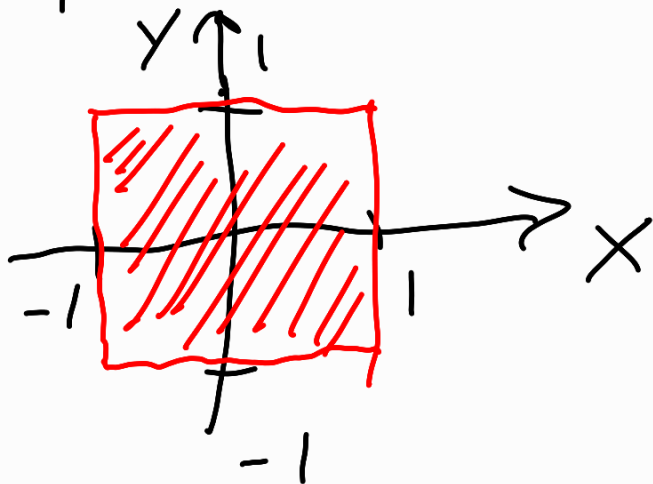
$$\Rightarrow \left\{ \begin{array}{l} 0 = x(1+y) \\ y = -0.5x^2 \end{array} \right\}$$

$$x=0 \quad \text{or} \quad y=-1$$

If  $x=0$ : Then  $y=0 \Rightarrow f=\underline{9}$

If  $y=-1$ : Then  $f = x^2 + x^2y + y^2 + 9 = x^2 - x^2 + (-1)^2 + 9 = \underline{10}$

Step 2: Check the boundary points of  $D$  by graphing



So we need to check  $y=\pm 1, x=\pm 1$

$$y=-1: f = x^2 - x^2 + y^2 + 9 = \underline{10}$$

$$y=1: f = 10 + 2x^2 \in \underline{[10, 12]}$$

$$x=\pm 1: f = y^2 + 9 + y + 1 = y^2 + y + 10$$



and  $-1 \leq y \leq 1$ .

$y = -1 \Rightarrow f = \underline{10}$ ,  $y = 1 \Rightarrow f = \underline{12}$ ,

and also solve  $0 = f' = 2y + 1$   
 $\Rightarrow y = -\frac{1}{2}$ , at which  $f = \underline{9.75}$

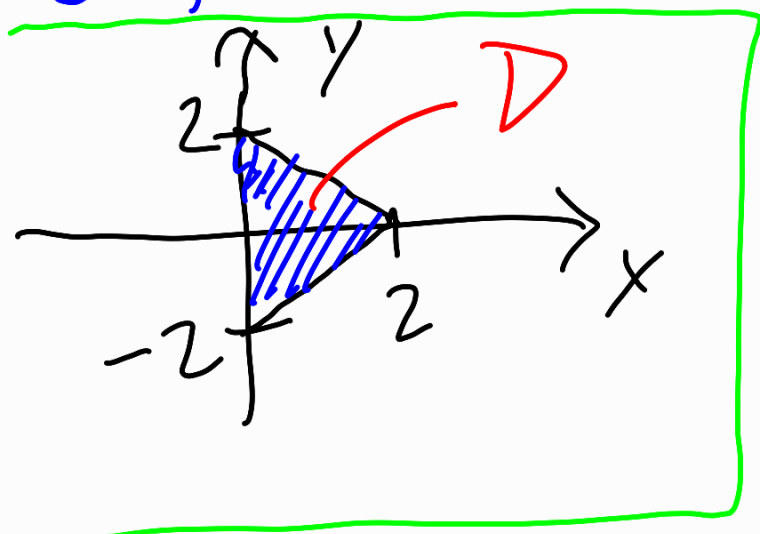
Step 3: Compare all values.

Lowest is 9, largest 12,

so  $\min = 9$ ,  $\max = 12$ .

(b)  $f = x^2 + y^2 - 2x$  where  $D$  is triangle w/ vertices  $(2, 0), (0, \pm 2)$

Step 1! Graph  $D$  & solve  $\nabla f = 0$



$$0 = \nabla f = \langle 2x-2, 2y \rangle$$

$$\Rightarrow \begin{cases} 2x=2 \\ 2y=0 \end{cases} \Rightarrow \begin{matrix} x=1 \\ y=0, \end{matrix}$$

$$\text{and } f(1,0) = \underline{-1}.$$

Step 2: Check perimeter.

The sides are  $x=0$ ,  $y=x-2$ ,

$$\text{and } y=2-x.$$

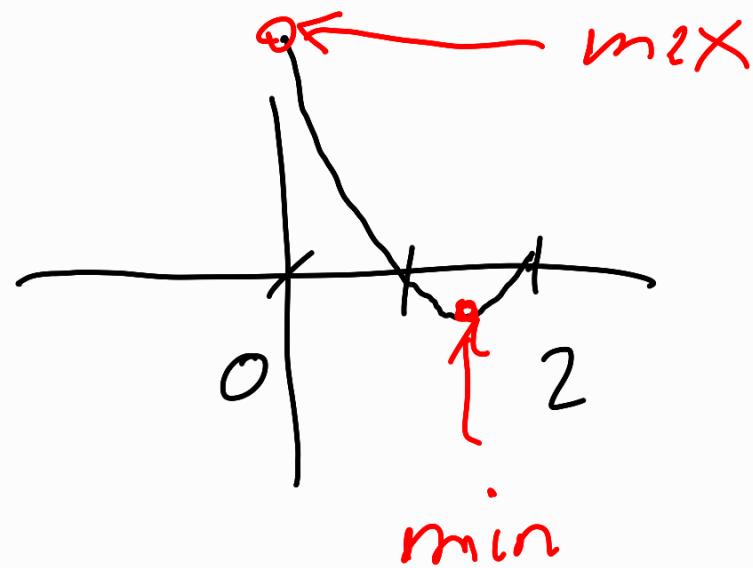
$$\text{If } x=0: f = y^2 \in \underline{[0,4]}$$

since  $-2 \leq y \leq 2$  on side

$$\text{If } y = x-2 \text{ or } 2-x. \text{ Then either case, } y^2 = (x-2)^2,$$

$$\text{so } f = x^2 - 2x + (x-2)^2 = 2x^2 - 6x + 4 = 2(x^2 - 3x + 2)$$

$$= 2(x-1)(x-2), \text{ and } 0 \leq x \leq 2.$$



By visual inspection from graphing,

max at  $x=0 \Rightarrow \underline{4}$ , min at  $x=1.5 \Rightarrow \underline{-\frac{1}{2}}$ .

Step 3: compare values.

$$\text{min} = -1, \text{ max} = 4.$$

4:  $f(x,y) = 48x + 96y - x^2 - 2xy - 9y^2$   
models profit/month where  $x$  is  
tennis balls (1000s),  $y$  is # ad  
hours purchased. Can produce at

most 50,000 balls/month & purchase  
at most 25 hours of ads. Find  
 $(x, y)$  that maximizes profit.

Step 1: Convert word prob to pure math

Goal: maximize  $f$ . Constraints on  
tennis balls  $\Rightarrow 0 \leq x \leq 50$ , also  
ads constraint  $\Rightarrow 0 \leq y \leq 25$ . So  
maximize  $f$  on  $[0, 50] \times [0, 25]$ .

Step 2: Solve  $\nabla f = 0$  for interior  
to find candidate max's in interior

$$0 = \nabla f = \langle 48 - 2x - 2y, 96 - 2x - 18y \rangle$$

$$\Rightarrow \begin{cases} x + y = 24 & (1) \\ x + 9y = 48 & (2) \end{cases}$$

Subtract (1) from (2) to get

$$24 = (x + 9y) - (x + y) = 8y \Rightarrow y = 3$$

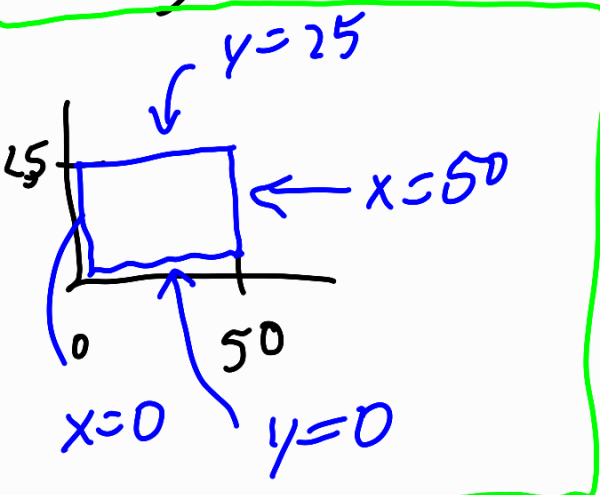
$$\Rightarrow x = 21 \Rightarrow (21, 3) \text{ only solution,}$$

and  $f(21, 3) = 1008 + 288 - 441 - 126 - 81 = 855 - 207 = \underline{648}$

$$\begin{array}{r} 48 \\ \cdot 21 \\ \hline 48 \\ 960 \\ \hline 1008 \end{array}$$

$$\begin{array}{r} 1296 \\ - 441 \\ \hline 855 \end{array}$$

Step 3: Check perimeter of the region  $[0, 50] \times [0, 25]$ .



We have to check  $x=0$ ,  $y=0$ ,  $y=25$ ,  $x=50$ .

$x=0$ :  $f = 96y - 9y^2$   
for  $0 \leq y \leq 25$ .

$$f(0) = \underline{0}, \quad f(25) = 25(96 - 9 \cdot 25) = 25(96 - 225) = -25 \cdot 229 = \underline{-5725},$$

Also check  $0 = f' = 96 - 18y$   
 $\Rightarrow y = 6$ , and  $f(6) =$

$$\begin{array}{r} 25 \\ \cdot 23 \\ \hline 75 \\ 500 \\ \hline 575 \end{array}$$

$$5750$$

$$6(96 - 9 \cdot 6) = 6 \cdot 48 = \underline{288}$$

$y=0$ :  $f = 48x - x^2$ .  $f(0) = \underline{0}$ ,  
 $f(50) = 50(48 - 50) = \underline{-100}$ , also  
 check  $0 = f' = 48 - 2x \Rightarrow x = 24$ ,  
 and  $f(24) = 24(48 - 24) = \underline{576}$

$y=25$  or  $x=50$ : Note we have  
 already checked  $(0, 25)$ ,  $(50, 0)$ .  
 So we only need to check  $(50, 25)$   
 and when  $f' = 0$ .

$$f(50, 25) = 2400 + 2400 - 2500 - 2500 \\ - 5625 = -200 - 5625 = \underline{-5825}.$$

$$\begin{array}{r} 24 \\ 625 \\ 9 \\ \hline 5625 \end{array}$$

If  $y=25$ :  $f = 48x - x^2 - 50x$   
 $+ 2500 - 5625 = -2x - x^2 - 3125$   
 $= -(x^2 + 2x + 3125) \leq -3125 <$

0, so no potential maxima in  
 this case.

If  $x=50$ :  $f = 96y - 100y - 9y^2 + 50(48-50) = -4y - 9y^2 - 100 = -(9y^2 + 4y + 100) \leq -100 < 0$ , so no potential maxima in this case.

Step 4: Look at all values found & where they occur.

Maximum is 648 at (21, 3),

so max profit is \$648,000.

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1: Find critical points, use 2nd derivative test to determine if local max, local min, or saddle

$$(2) f = 4x^2 + 9y^2 + 8x - 36y + 24$$

Step 1: Solve  $\nabla f = 0$  for candidates

$$0 = \nabla f = \langle 8x + 8, 18y - 36 \rangle$$



$$\Rightarrow \begin{cases} 8x+8=0 \\ 18y-36=0 \end{cases} \Rightarrow \begin{matrix} x=-1 \\ y=2 \end{matrix}, \text{ so}$$

$(-1, 2)$  is the only crit. point

Step 2: Apply 2nd derivative test.

Recall: For the test,  $H := f_{xx}f_{yy} - f_{xy}^2$  is computed and we check sign of  $H$ .

$$f_{xx} = 8, f_{yy} = 18, f_{xy} = 0, \text{ so}$$

$$H = 8 \cdot 18 - 0 = 144 > 0 \text{ for}$$

all points  $(x, y)$ . And  $f_{xx} > 0$ ,

so  $f$  has local min @  $(-1, 2)$ .

$$(b) g = (y-2)x^2 - y^2$$

Step 1: Solve  $\nabla g = 0$ :

$$\vec{0} = \nabla g = \langle 2x(y-2), x^2 - 2y \rangle$$

$$\Rightarrow \begin{cases} x(y-2) = 0 \\ x^2 = 2y \end{cases}$$

So either  $x=0$  or  $y=2$



If  $x=0$ :  $2y=0 \Rightarrow y=0 \Rightarrow \underline{(0,0)}$

If  $y=2$ :  $x^2=4 \Rightarrow x=\pm 2 \Rightarrow \underline{(\pm 2, 2)}$

Step 2: Apply 2nd deriv. test

$$g_{xx} = 2(y-2), \quad g_{yy} = -2,$$

$$g_{xy} = 2x, \text{ so}$$

$$D = -4(y-2) - (2x)^2 = -4[(y-2) + x^2].$$

$$D(0,0) = 8 > 0, \quad g_{xx}(0,0) = -4$$

$< 0$ , so  $\boxed{(0,0) \text{ local max.}}$

$$D(\pm 2, 2) = -4((\pm 2)^2) = -16 < 0,$$

so  $\boxed{(\pm 2, 2) \text{ are saddles.}}$

HW Q4: Extrema  $f = x^4 - 2x^2 + y^3 - 3y$

$$0 = \nabla f = \langle 4x^3 - 4x, 3y^2 - 3 \rangle$$

$$\Rightarrow x = 0, \pm 1, \quad y = \pm 1.$$

$$f_{xx}f_{yy} - f_{xy}^2 = (12x^2 - 4)(6y) - 0^2 = 24(3x^2 - 1)y.$$

$$D(0, 1) = -24 < 0 \Rightarrow (0, 1) \text{ saddle}$$

$$D(0, -1) = 24 > 0, f_{xx}(0, -1) = -4 < 0 \Rightarrow (0, -1) \text{ local max}$$

$$D(\pm 1, 1) = 48 > 0, f_{xx}(\pm 1, 1) = 8 > 0 \Rightarrow (\pm 1, 1) \text{ local min}$$

$$D(\pm 1, -1) = -48 < 0 \Rightarrow (\pm 1, -1) \text{ saddle}$$

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